

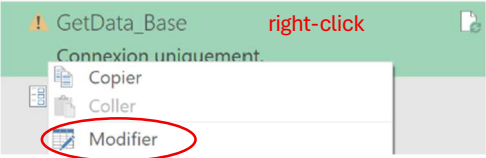
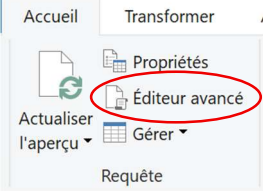
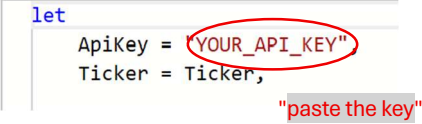
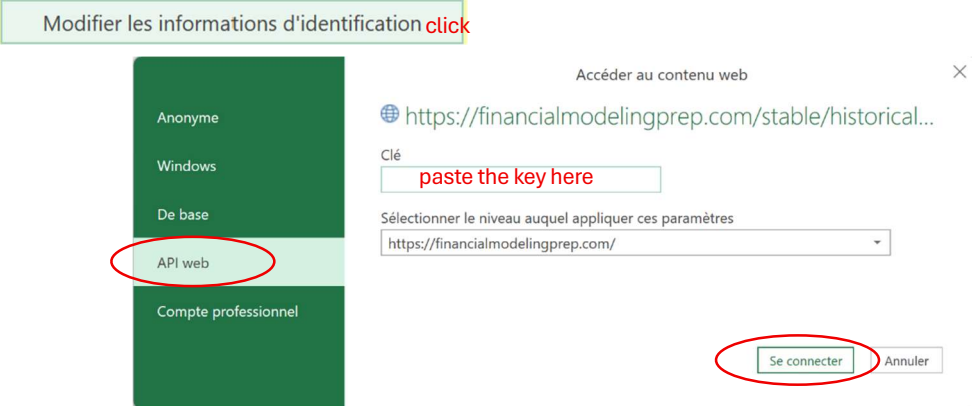
How to Use

Version 2.2 | Last Updated: 09/29/2026 | by Ilona Gavoille

Requirements

To use this portfolio optimization model, you will need:

- **Microsoft Excel Desktop**
- **Solver Add-in** enabled in Excel
 - <https://support.microsoft.com/en-us/excel/load-the-solver-add-in-in-excel>
- **Internet connection** to retrieve financial market data
- **Personal API key** from the Financial Modeling Prep API:
 - Create an account to get an API key on <https://site.financialmodelingprep.com/>
 - Insert the key in Excel Query:

- 1. Excel ribbon with 'Données' tab selected.
- 2. Right-click context menu for 'GetData_Base' with 'Modifier' selected.
- 3. Excel ribbon with 'Éditeur avancé' selected.
- 4. Excel query editor showing API key input: `let
ApiKey = "YOUR_API_KEY",
Ticker = Ticker,` with a red box around "YOUR_API_KEY" and the text "paste the key".
- 5. Login dialog box for Financial Modeling Prep. The 'API web' option is selected. The 'Clé' field contains 'paste the key here'. The 'Se connecter' button is highlighted.

1. Enable macros

When opening the file with format .xlsm file, **enable macros** if asked.

The workbook opens on the Dashboard, which is the main user interface.

2. Refresh the market data

Click **REFRESH**.

Note: The workbook initially contains data from the previous use. Always refresh the data before generating a new set of portfolios.

3. Set the optimization parameters

Once the data has been refreshed, click **OPTIMIZE PORTFOLIO** on the Dashboard.

Follow the guidelines and enter in the yellow cells:

- **Minimum weight** for each asset
- **Maximum weight** for each asset
- **Risk-free rate** : choose a rate appropriate for your analysis, such as the 10-Year U.S. Treasury yield, T-Bill rate, or another relevant risk-free benchmark.

These parameters define the constraints used by the optimization model.

4. Generate the portfolios

Once the parameters have been entered, click **GENERATE**.

The model automatically runs Excel Solver and generates:

- **12 portfolios** along the efficient frontier
- **1 Maximum Sharpe portfolio**
- **1 Minimum Variance portfolio**

The optimization may take a few moments depending on the number of constraints and the computer used.

The portfolio details, efficient frontier and Capital Allocation Line are displayed in the 03_Efficient Frontier sheet.

5. Explore the results

Return to the **Dashboard**. Use the **Selected Portfolio** dropdown to choose between the Portfolio 1 to Portfolio 12, Maximum Sharpe Portfolio or Minimum Variance Portfolio.

The Dashboard then updates dynamically according to the selected portfolio.